

How to Select a Project Topic Report

Introduction and Justification of Topic

The detection of deepfakes—synthetic audio, video, or image content generated by artificial intelligence—has rapidly become one of the most pressing challenges in computer science and society. The technology’s capacity to manipulate reality threatens public trust, digital integrity, and democratic institutions. Recent studies highlight that while public creation of deepfakes remains limited, concern about their misuse is overwhelming. Sippy et al. (2024) report that over 90% of UK adults are worried about deepfakes’ potential to spread misinformation, harass individuals, and undermine factual communication. These fears are justified by the exponential improvement in AI generative models such as GANs and diffusion architectures, which now produce highly realistic fake media indistinguishable to the human eye.

From a Software Engineering perspective, developing a robust, usable, and context-aware deepfake detection and verification framework directly addresses a critical need: to integrate AI-driven analysis with reliable software systems that can identify manipulated media in real-world conditions. It is therefore an academically and societally valid topic for a final-year project, merging AI model development, system integration, and user-facing application design.

Developments and Research Trends (2019–2024)

Deepfake detection research has advanced significantly in the last four years, evolving from simple image classifiers to multi-modal and explainable AI systems.

Gong and Li (2024) provide a comprehensive overview of this trajectory, identifying four major categories of detection methods: (1) CNN-based models; (2) semi-supervised consistency learning; (3) transformer-based architectures; and (4) biological signal-based detection. Their work demonstrates that while several detectors achieve high accuracy within specific datasets (often above 90%), they fail to generalize to unseen or higher-quality fakes—a recurring challenge echoed by Kwon et al. (2021) and Zi et al. (2020). The KoDF and WildDeepfake datasets, for example, revealed that models trained on synthetic benchmarks like FaceForensics++ perform poorly in real-world conditions, dropping by up to 20 percentage points in accuracy.

Lyu (2020) underscores that deepfake detection remains a “technological arms race.” He identifies three principal manipulation types—head puppetry, face swapping, and lip syncing—and calls for explainable detectors capable of localizing manipulations rather than merely classifying them. Similarly, Heidari et al. (2023) highlight the importance of interdisciplinary approaches, noting that deepfakes now extend beyond visuals into audio and hybrid modalities, necessitating multi-modal detection systems that combine visual, acoustic, and physiological cues.

Khalid et al. (2021) advance this idea with the FakeAVCeleb dataset, the first large-scale audio–video multimodal benchmark, demonstrating that dual-stream neural networks outperform single-modality models (F1 = 94.3 vs 88.5 for video-only).

At the algorithmic level, CNNs remain the dominant architecture, as reaffirmed by Ahmed et al. (2022), whose comparative study of eight convolutional models achieved up to 97% accuracy using VGG-Face on static deepfake images. Yet, Pan et al. (2020) show that ensemble CNNs (e.g., Xception + MobileNet) trained on FaceForensics++ still struggle with unseen manipulations such as NeuralTextures or StyleGAN fakes.

These works collectively reveal a research community focused on improving generalization, multi-modal fusion, explainability, and dataset realism.

Sub-Problems Identified in the Literature

Across these studies, four recurring subproblems emerge:

1. Generalization and Dataset Bias

Detectors perform well on specific datasets but fail on new fake types due to overfitting (Gong and Li, 2024; Kwon et al., 2021; Zi et al., 2020). Cross-dataset robustness remains the most cited weakness.

2. Multi-Modality and Synchronization

Traditional detectors focus solely on images or video frames. New challenges involve matching audio–visual consistency and detecting multimodal deepfakes (Khalid et al., 2021).

3. Explainability and Localized Detection

Lyu (2020) emphasizes the need for interpretable, localized results—identifying where manipulation occurs rather than issuing a single “fake” label.

4. Real-Time and Practical Deployment

Most deepfake detectors are computationally intensive and unsuitable for mobile or live environments (Heidari et al., 2023). There is a clear gap for lightweight yet accurate systems deployable at scale.

Proposed Approach and Testbed

This project aims to develop a modular detection framework—not just a single model—integrating AI algorithms with a usable software system. The testbed will combine pre-trained CNN and transformer architectures within a Python-based framework.

- **Programming Framework:** Python 3.12 using TensorFlow and PyTorch.
- **Dataset Sources:** Publicly available datasets such as FaceForensics++ (Rössler et al., 2019), FakeAVCeleb (Khalid et al., 2021), and WildDeepfake (Zi et al., 2020).
- **Model Design:** Use transfer learning from VGG-Face and Xception models for visual detection, fused with an audio-embedding branch (MFCC spectrogram features).
- **System Architecture:** A modular “backend + API + UI” design, allowing the framework to be reused as a library or extended into a full application.
- **Platform & Tools:** Google Colab / University GPU servers for training; Flask API for integration; React-based web front-end for demonstration.

The framework will function as a testbed for experimentation, enabling side-by-side comparison of model architectures and real-time verification of video uploads.

Evaluation Path and Metrics

Evaluation will involve both technical and usability assessments, in line with research practices from Ahmed et al. (2022) and Pan et al. (2020).

- **Quantitative Metrics:** Accuracy, Precision, Recall, F1 Score, and AUC (Area Under Curve) on benchmark datasets.
- **Cross-Dataset Evaluation:** Train on one dataset (e.g., KoDF) and test on another (e.g., FakeAVCeleb or WildDeepfake) to assess generalization.
- **Efficiency Metrics:** Processing latency (ms/frame) and model size (MB) for real-time viability.
- **User Study Metrics:** Usability via System Usability Scale (SUS) and perceived trust in results (based on Sippy et al., 2024’s finding that 90% of users worry about authenticity).
- **Explainability Assessment:** Visual heatmaps (Grad-CAM) showing manipulated regions for interpretability.

The project's success will be measured by:

1. $\geq$ Ninety percent accuracy on benchmark datasets,
2. Cross-dataset robustness $> 80\%$,
3. Detection latency < 300 ms per frame, and
4. Positive SUS $> 75/100$ from user testing.

Conclusion

The deepfake phenomenon poses technical, ethical, and societal challenges that require interdisciplinary solutions. As demonstrated across recent studies—from the WildDeepfake and FakeAVCeleb datasets to CNN- and transformer-based models—researchers are striving to improve generalization, transparency, and real-world readiness. However, there remains a clear research and engineering gap for a modular, lightweight, context-aware detection framework that integrates these insights into a deployable software system.

References

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